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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Jr.SUPER60_NUCLEUS & STERLING BT WTA-24/ WTA-19
Time: 09:00AM to 12:00PM JEE-ADV_(2020-P2)

Date: 29-10-2023
Max. Marks: 198

2020_PAPER-II

29-10-2023_Jr.Super60_NUCLEUS & STERLING BT_Jee-Adv(2020-P2)_WTA-24&WTA-19_Syllabus

PHYSICS

: Units and dimensional analysis of physical quantities taught till now, Significant figures, Rounding off of numbers, Accuracy, Precision (All rules pertaining to above topics only as per NCERT text book), Error Analysis of Physical quantities, Vernier callipers, Screw gauge

CHEMISTRY

: GOC II :Reaction intermediates : Homolytic and heterolytic cleavage of bonds, Carbocations: Generation, Structure and stability, Rearrangement of carbocations., Carbocations: Generation, Structure and stability, Rearrangement of carbocations., Carbanions and free radicals: Generation, Structure and Stability, Benzyne, carbenes, nitrenes and radical ions: Generation, Structure and Stability

MATHEMATICS

: Differentiation: from First Principle, Basic Rules of Differentiation, Logarithmic & Parametric Differentiation, Differentiation of Implicit Functions, Differentiation of composite functions, Differentiation using substitution, Derivative of a function w.r.t another function, Summation of some Series using Derivative, derivative of inverse of a function

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2020-P2-Model**

Time: 3:00Hour's

IMPORTANT INSTRUCTIONS

Max Marks: 198

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 7 – 12)	Questions with Multiple Correct Choice +1 partial marks	4	-2	6	24
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	4	0	6	24
Total				18	66

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 25 – 30)	Questions with Multiple Correct Choice +1 partial marks	4	-2	6	24
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	4	0	6	24
Total				18	66

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 43 – 48)	Questions with Multiple Correct Choice +1 partial marks	4	-2	6	24
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	4	0	6	24
Total				18	66

Sec: Jr.Super60_NUCLEUS & STERLING BT

Space for rough work

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Educational Institutions**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023****JEE MAIN
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RANK**1****JEE Advanced
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MAY 2023
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6th-12th Class
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RANK**1****NEET
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PHYSICS

Max Marks: 66

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct integer is entered;

Zero Marks : 0 If the question is unanswered;

Negative Marks : -1 In all other cases

- Force on a particle in one-dimensional motion is given by $F = Av + Bt + \frac{Cx}{At + D}$ where F is force, v is speed, t is time, x is position and A, B, C and D are constants. Dimensions of C are $[M^p L^q T^r]$. The value of $(p - q - r)$ is
- The value of x in the formula $Y = \frac{2mgl^x}{5bt^3e}$ is (where ' m ' is the mass, ' g ' is acceleration due to gravity, ' l ' is the length, ' b ' is the breadth, ' t ' is the thickness and ' e ' is the extension and Y is Young's modulus)
- If force F , power P and impulse J are taken as the fundamental quantities in a system, then dimensional formula of length in this system is given by $[F^x P^y J^z]$. Find $x + y + z$
- If the velocity of light, acceleration due to gravity and atmospheric pressure are chosen as fundamental units, then the units of mass is $9n \times 10^{34} \text{ kg}$. Find n .
($C = 3 \times 10^8 \text{ ms}^{-1}$; $g = 10 \text{ ms}^{-2}$; $P_0 = 10^5 \text{ N / m}^2$)
- A person measures the depth of a well by measuring the time interval between dropping a stone and receiving the sound of impact with the water in the well. The error in his measurement of time is $\delta T = 0.01$ seconds and he measures the depth of the well to be $L = 20$ metres. Take the acceleration due to gravity $g = 10 \text{ ms}^{-2}$ and the velocity of sound is 300 ms^{-1} . Then $\frac{\delta L}{L} \times 100$, is closest to

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6. Velocity (v) and acceleration (a) in two systems of units 1 and 2 are related as

$v_2 = \frac{n}{m^2} v_1$ and $a_2 = \frac{a_1}{mn}$ respectively. Here m and n are constants. The relations for

distance and time in two systems respectively are $P \frac{n^3}{m^3} L_1 = L_2$ and $Q \frac{n^2}{m} T_1 = T_2$.

Find $P + Q$.

SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

• This section contains SIX (06) questions.

• Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

• For each question, choose the option(s) corresponding to (all) the correct answer(s).

• Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks** +3 If all the four options are correct but ONLY three options are chosen;

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

7. The potential energy of a particle as function of its position ' x ' is given by the

expression $U(x) = -\alpha x + \beta \sin \frac{x}{\gamma}$. Among the following a dimensionless combination

of the constants α, β and γ is:

- A) $\frac{\alpha}{\beta\gamma}$ B) $\frac{\alpha^2}{\beta\alpha}$ C) $\frac{\gamma}{\alpha\beta}$ D) $\frac{\alpha\gamma}{\beta}$

8. In an experiment to determine the acceleration due to gravity g , the formula used for the

time period of a periodic motion is $T = 2\pi \sqrt{\frac{7(R-r)}{5g}}$. The values of R and r are

measured to be (60 ± 1) mm and (10 ± 1) mm, respectively. In five successive measurements, the time period is found to be 0.52s, 0.56s, 0.57s, 0.54s and 0.59s. The least count of the watch used for the measurement of time period is 0.01s. Which of the following statement(s) is (are) true?

- A) The error in the measurement of r is 10%
 B) The error in the measurement of T is 3.57 %
 C) The error in the measurement of T is 2%
 D) The error in the determined value of g is nearly 11%



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9. Consider the efficiency of carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta} \log_e \left(\frac{\beta x}{kT} \right)$, where α and β are constants. If T is temperature, k is Boltzmann constant, θ is angular displacement and x has the dimensions of length. Then, choose the correct option:
- A) Dimensions of β is same as that of force.
B) Dimensions of $\alpha^{-1}x$ is same as that of energy.
C) Dimensions of $\eta^{-1} \sin\theta$ is same as that of $\alpha\beta$.
D) Dimensions of α is same as that of β
10. Choose the correct among the following
- A) 5.604 has four significant figures B) 0.045 has two significant figures
C) 343.000 has six significant figures D) 1.507×10^4 has four significant figures
11. Among the following limitations of dimensional analysis is
- A) If a physical quantity depends on more than three physical quantities, this method cannot derive the relationship
B) Using this method dimensionless constants in a relation cannot be found
C) It fails so derive the relations like $S = ut + \frac{1}{2}at^2$
D) This method fails to derive the relationships which involve trigonometric, logarithmic and exponential functions
12. Among the following identify the set with same dimensions
- A) Velocity gradient, Frequency, Angular velocity
B) Angular momentum, Impulse, moment of force
C) Work, moment of force, potential energy
D) Radius of gyration, wavelength, displacement





SECTION – III (NUMERICAL VALUE TYPE)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 In all other cases..

13. The units of force, velocity and energy are 100 dyne, 10 cm s^{-1} and 500 erg respectively. The units of mass, length and time are a grams, b centimetres and c seconds. Find $a - b + c$.
14. The period of oscillation of a simple pendulum is given by $T = 2\pi \sqrt{\frac{l}{g}}$, where l is about 100 cm and is known to have 1 mm accuracy. The period is about 2 s. The time of 100 oscillations is measured by a stop watch of least count 0.1 s. The percentage error in g is m and the percentage error in time period is n . Find $m + n$.
15. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has an error of 2%, then the percentage error in the density is nearly.
16. In an experiment with Vernier calipers of least count 0.1 mm, when two jaws are joined together the zero of Vernier scale lies to the right zero of the main scale and 6th division of Vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between 3.2 cm and 3.3 cm marks on main scale, and 4th division of vernier scale coincides with the main scale division. The diameter of bob in cm is measured as
17. A screw gauge of pitch 0.5 mm is used to measure the diameter of uniform wire of length 6.8 cm, the main scale reading is 1.5 mm and circular scale reading is 7. The calculated curved surface area of wire in cm^2 to appropriate significant figures is :
 [Screw gauge has 50 divisions on its circular scale and $\pi = 3.14$]
18. A physical quantity z depends on four observables a, b, c and d , as $Z = \frac{a^2 b^3}{\sqrt{c} d^3}$. The percentages of error in the measurement of a, b, c and d are 2 %, 1.5%, 4% and 2.5% respectively . The percentage of error in z is :

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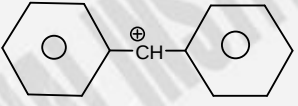
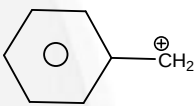
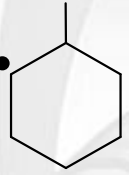
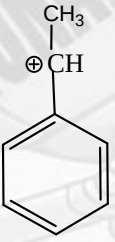
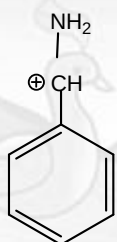
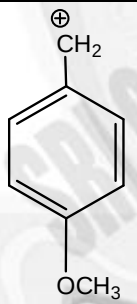
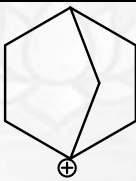
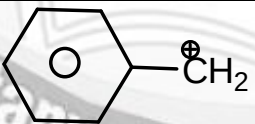
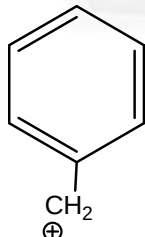
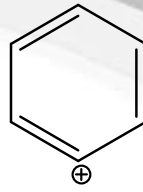
CHEMISTRY

Max Marks: 66

SECTION-I
(INTEGER ANSWER TYPE)

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Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

19. How many of the following pairs, first intermediate is more stable than the second one?

1)	$CH_3 - CH = \overset{\oplus}{C}H$, $CH_2 = CH - \overset{\oplus}{C}H_2$	2)	 , 
3)	 , 	4)	 , 
5)	 , 	6)	 , 
7)	$CH_2 = CH - \overset{\oplus}{C}H_2$, 	8)	$CF_3 - \overset{\ominus}{C}H_2$, $CF_3 - \overset{\ominus}{C}H - CF_3$
9)	 , 		

Sec: Jr.Super60_NUCLEUS & STERLING BT

Space for rough work

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THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

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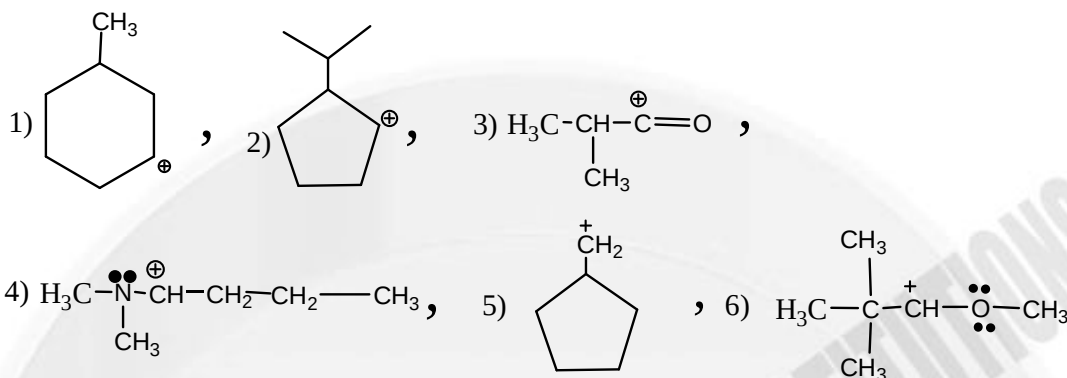
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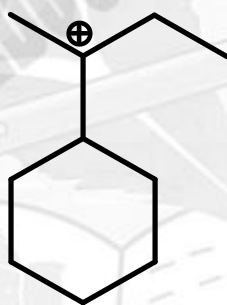
20. How many among the given will undergo carbocations rearrangement?



21. Formal charge on singlet carbene is X

Formal charge on triplet carbene is Y then $\frac{X+Y}{2}$ is

22. The total number of contributing structures showing hyperconjugation (involving C-H bond) for the following carbocation is (including the given structure)



23. Difference in the number of sp^2 -hybridized carbons to the sp -hybridized carbons in Benzene is
24. How many of the following statements are correct
- Carbocations act as electrophiles
 - Free radicals are paramagnetic
 - Carbenes have incomplete octet
 - Allyl carbocation is more stable than propyl carbocation
 - Carbanions stability : $3^\circ > 2^\circ > 1^\circ > \text{methyl}$
 - Benzyl carbocation is less stable than allyl carbocation
 - Isopropyl free radical is less stable than 3° -butyl free radical
 - Dichlorocarbene has 2 lone pairs
 - Carbanions are formed by heterolysis of a covalent bond.





SECTION – II

(ONE OR MORE CORRECT ANSWER TYPE)

• This section contains SIX (06) questions.

• Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

• For each question, choose the option(s) corresponding to (all) the correct answer(s).

• Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks** +3 If all the four options are correct but ONLY three options are chosen;

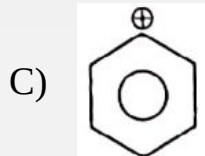
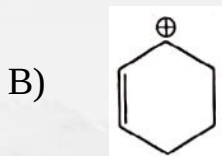
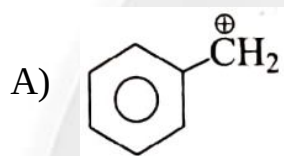
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

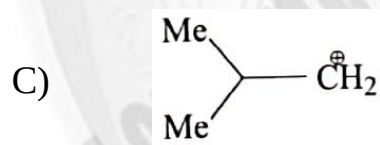
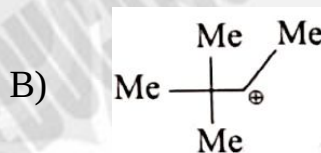
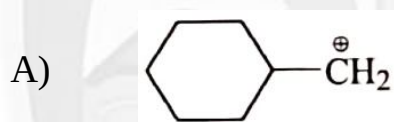
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

25. Which of the following carbocations is/are resonance stabilized?



26. In which of the following ΔG decreases if there can be some intramolecular rearrangement



27. Select the correct statements :

A) Singlet $\bullet\text{CCl}_2$ is more stable than singlet $\bullet\text{CF}_2$

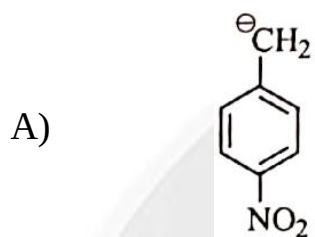
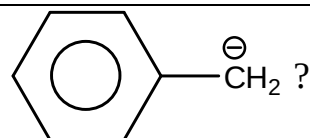
B) Triplet $\bullet\text{CH}_2$ is more stable than singlet $\bullet\text{CH}_2$

C) Singlet $\bullet\text{CH}_2$ has bent shape

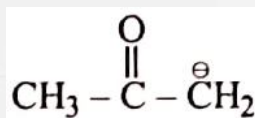
D) Triplet nitrene is more stable than singlet nitrene



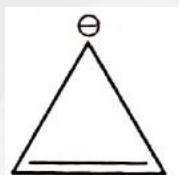
28. Among following anions which are more stable than



B)



C)

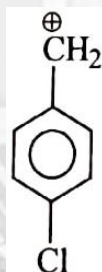


D)

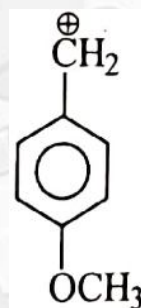


29. Most stable carbocation is :

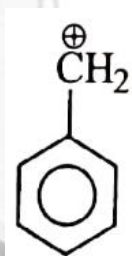
A)



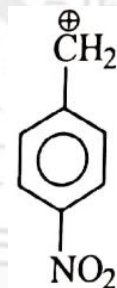
B)



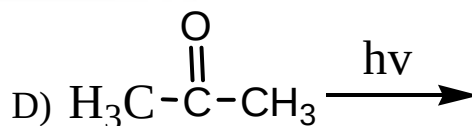
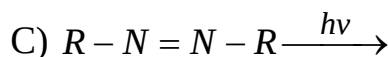
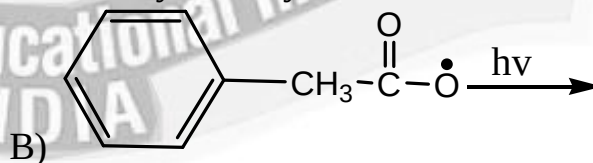
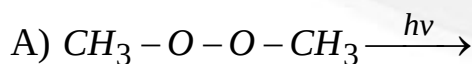
C)



D)



30. In which case(s), free radicals can be formed by homolytic fission?

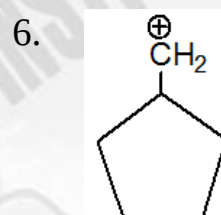
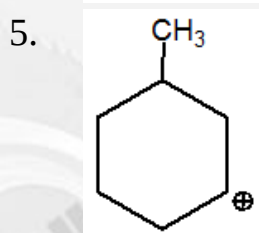
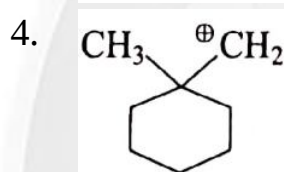
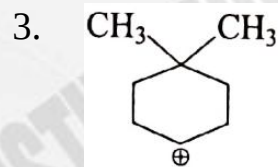
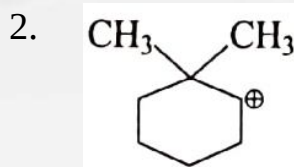
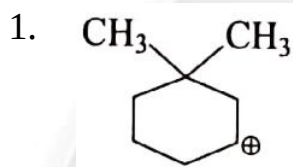




SECTION – III (NUMERICAL VALUE TYPE)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If **ONLY** the correct numerical value is entered;
Zero Marks: 0 In all other cases..

31. How many of the following carbocations is likely to rearrange ?



32.

Intermediate species	Number of Hybrid orbitals
Carbocation	A
Carbanion	B
Free Radical	C
Singlet Carbene	D
Triplet Nitrene	E

Then the value of $\frac{A + B + C - D - E}{3}$



33.

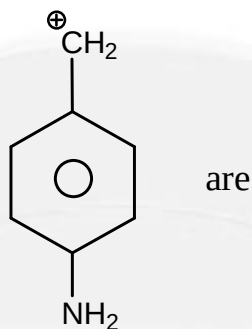
Number of primary hydrogens in X are A
 Number of secondary hydrogens in X are B
 Number of tertiary hydrogens in X are C

Then $\left(\frac{A + B + C}{2}\right)$ is

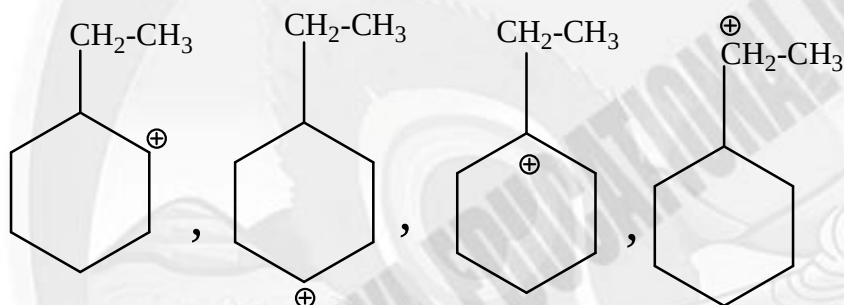




34. The number of π – bonds in most stable resonating structure of (excluding the given structure)



35. The number of secondary hydrogens in the most stable carbocation among the following



36. The number of π – bonds present in trityl cation is/are

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**MATHEMATICS****Max Marks: 66****SECTION-I
(INTEGER ANSWER TYPE)**

- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

37. Let $y = \frac{2^{\log_{2^{1/4}} x} - 3^{\log_{27}(x^2+1)^3} - 2x}{7^{4\log_{49} x} - x - 1}$ and $\frac{dy}{dx} = ax + b$, then the value of $a + b$ is _____.

38. If $f : R \rightarrow R$ satisfies $|f(x) - f(y)| \leq |x - y|^3$ and $f(4) = 192$ then $\frac{f(7)}{24}$ is equal to _____

39. If $|x| < 1$ and $\frac{1-2x}{1-x+x^2} + \frac{2x-4x^3}{1-x^2+x^4} + \frac{4x^3-8x^7}{1-x^4+x^8} + \dots = \frac{1+ax}{1+bx+cx^2}$ then $a + b - c$ is _____

40. Let $g(x) = \begin{cases} \frac{x^2 + x \tan x - x \tan 2x}{ax + \tan x - \tan 3x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ If $g^1(0)$ exist and is equal to non-zero value b then $52\left(\frac{b}{a}\right)$ is equal to _____

41. If $\lim_{t \rightarrow x} \frac{e^t f(x) - e^x f(t)}{(t-x)(f(x))^2} = 2$ and $f(0) = \frac{1}{2}$, then the value of $f'(0)$ is _____

42. A function is represented parametrically by the equation $x = \frac{1+t}{t^3}; y = \frac{3}{2t^2} + \frac{2}{t}$. Then the value of $4 \left| \frac{dy}{dx} - x \left(\frac{dy}{dx} \right)^3 \right|$ is _____

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SECTION - II

(ONE OR MORE CORRECT ANSWER TYPE)

• This section contains SIX (06) questions.

• Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

• For each question, choose the option(s) corresponding to (all) the correct answer(s).

• Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks** +3 If all the four options are correct but ONLY three options are chosen;

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

43. If $e^{\sin(x^2+y^2)} = \tan \frac{y^2}{4} + \sin^{-1} x$, then $y'(0)$ can be

- A) $\frac{1}{3\sqrt{\pi}}$ B) $-\frac{1}{3\sqrt{\pi}}$ C) $-\frac{1}{5\sqrt{\pi}}$ D) $-\frac{1}{3\sqrt{5\pi}}$

44. If $y = \frac{\sqrt{1+t^2} - \sqrt{1-t^2}}{\sqrt{1+t^2} + \sqrt{1-t^2}}$ and $x = \sqrt{1-t^4}$ then the value of $\frac{dy}{dx}$ is/are

- A) $\frac{-1}{t^2(1+\sqrt{1-t^4})}$ B) $\frac{(\sqrt{1-t^4}-1)}{t^6}$

- C) $\frac{1}{t^2(1+\sqrt{1-t^4})}$ D) $\frac{1-\sqrt{1-t^4}}{t^6}$

45. Let $y = \sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$ then which is/are correct ?

- A) $\frac{dy}{dx} = \frac{1}{2y-1}$ B) $\frac{dy}{dx} = \frac{x}{x+2y}$ C) $\frac{dy}{dx} = \frac{1}{\sqrt{1+4x}}$ D) $\frac{dy}{dx} = \frac{y}{2x+y}$

46. If $f(\theta) = \tan\left(\sin^{-1}\sqrt{\frac{2}{3+\cos 2\theta}}\right)$, then which of the following option(s) is/are correct?

- A) $f\left(\frac{\pi}{4}\right) = 1$ B) $f\left(\frac{\pi}{4}\right) = \sqrt{2}$
- C) $\frac{d(f(\theta))}{d(\cos \theta)}$ at $\theta = \frac{\pi}{4}$ is -2 D) $f'\left(\frac{\pi}{4}\right) = \sqrt{2}$

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47. Let $f: R \rightarrow R, g: R \rightarrow R$ and $h: R \rightarrow R$ be differentiable functions such that $f(x) = x^3 + 3x + 2, g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in R$. Then which of the following option(s) is/are correct?
- A) $g'(2) = \frac{1}{15}$ B) $h'(1) = 666$ C) $h(0) = 16$ D) $h(g(3)) = 36$
48. If $y = x^{(\log x)^{\log(\log x)}}$ then which is/are correct ?
- A) $\frac{dy}{dx} = \frac{y}{x} \left((\ln x^{x-1}) + 2 \ln x \ln(\ln x) \right)$
- B) $\frac{dy}{dx} = \frac{y}{x} (\log x)^{\log(\log x)} (2 \log(\log x) + 1)$
- C) $\frac{dy}{dx} = \frac{y}{x \ln x} \left[(\ln x)^2 + 2 \ln(\ln x) \right]$
- D) $\frac{dy}{dx} = \frac{y}{x} \frac{\log y}{\log x} [2 \log(\log x) + 1]$

SECTION – III

(NUMERICAL VALUE TYPE)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
 - For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
 - Answer to each question will be evaluated according to the following marking scheme:
- Full Marks:** +4 If ONLY the correct numerical value is entered;
- Zero Marks:** 0 In all other cases..

49. If $y = \cos^{-1} \sqrt{\frac{\cos 3x}{\cos^3 x}}$ and $\frac{dy}{dx} = \sqrt{\frac{k}{\cos 2x + \cos 4x}}$, ($\sin x > 0$) then k is _____
50. Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then absolute value of $y^1(1)$ equals _____
51. If $f(x) = \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x$ then $f^1(\pi/4) =$ _____
52. If $f(x) = \sqrt{x + 2\sqrt{2x-4}} + \sqrt{x - 2\sqrt{2x-4}}$, then the value of $10 f'(102^+)$ is _____.
53. If $y = \frac{\sqrt[3]{1+3x} \sqrt[4]{1+4x} \sqrt[5]{1+5x}}{\sqrt[7]{1+7x} \sqrt[8]{1+8x}}$, then $y'(0)$ is equal to _____.
54. If $y = \frac{a + bx^{3/2}}{x^{5/4}}$ and $y' = 0$ at $x = 5$, then the value of $\frac{3a^2}{2b^2}$ is _____.

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